STAT 332 Sampling and Experimental Design

University of Waterloo - Fall 2025

8th October 2025

Instructor: Samuel Wong

LaTeX : Xing Liu

Contents

1 Sampling

Lecture 1, 2025/09/03

1.1 Intro & Survey Terminology

Assignments are there to help you prepare for exams!

Why study survey sampling?

- Obtain data for a question of interest from a population.
- Cannot take a census.
- Finite time and resources.

Definition 1.1 (**Target Population**).
The complete collection of individuals we wish to study and draw conclusions about. Denoted by $U = \{1, 2, \dots, N\}$.

Definition 1.2 (**Sampled (Study) Population**).
The collection of individuals that could be included by our sampling design.

Definition 1.3 (**Sampling Unit**).
The object being sampled.

Definition 1.4 (**Frame**).
The list of all sampling units.

Definition 1.5 (**Observation Unit**).
The object on which we take measurements.

Note. Often, this is the person we ask questions to.

Three main steps at which error and bias can be introduced in the conducting of a survey:

- Constructing the frame (coverage error or study error)

- Obtaining the sample/respondents (sampling bias/error, non-response error)
- Measurement of the response (measurement error)

Definition 1.6 (**Sampling Design (Sampling Protocol)**).
A **sampling design** (or sampling protocol) is the mechanism by which we select our samples.

Note. **Sampling error** is the estimation error due to sampling variability.

Definition 1.7 (**Selection Bias**).
Differences between the target population and the effective sampled population.

Note. Ideally, target population = population that can be effectively sampled.

Sources of selection bias:

- Coverage: e.g., can be difficult to construct a complete frame
- Sampling: e.g., improper sampling design
- Non-response: people don't respond or are not reachable

Definition 1.8 (**Measurement Error**).
The measured values of the attributes on the observation units are different from the true values.

Lecture 2, 2025/09/08

1.2 Probability-Based Sampling Design

Assume that frame is complete (ideal case). Assume also that $U = \{1, 2, \dots, N\}$. We denote the responses (corresponding values for attribute of interest) by $y_1, y_2, \dots, y_N$.

Population quantities:

- $\mu = \frac{1}{N} \sum_{i \in U} y_i$ (population mean)
- $\tau = \sum_{i \in U} y_i$ (population total)
- $\sigma^2 = \frac{1}{N-1} \sum_{i \in U} (y_i - \mu)^2 = \frac{1}{N-1} \left[\sum_{i \in U} y_i^2 - N\mu^2\right]$ (population variance b/c population is finite).

Note.

$$\begin{aligned}
\sigma^2 &= \frac{1}{N-1}\sum_{i\in U}(y_i-\mu)^2 \\
&= \frac{1}{N-1}\sum_{i\in U}(y_i^2-2\mu y_i+\mu^2) \\
&= \frac{1}{N-1}\left[\sum_{i\in U}y_i^2-2\mu\sum_{i\in U}y_i+N\mu^2\right] \\
&= \frac{1}{N-1}\left[\sum_{i\in U}y_i^2-2N\mu^2+N\mu^2\right] \\
&= \frac{1}{N-1}\left[\sum_{i\in U}y_i^2-N\mu^2\right].
\end{aligned}$$

Suppose we have a sample $s \subseteq U$, a subset of frame with sample size n.

<u>Estimates</u> of population quantities based on sample:

- $\hat{\mu} = \frac{1}{n}\sum_{i\in s} y_i$, sample mean estimates population mean
- $\hat{\tau} = \sum_{i\in s} y_i$, sample total estimates population total
- $\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i\in s}(y_i-\hat{\mu})^2$, sample variance estimates population variance

Note. As in STAT 231, we treat estimates as constant values based on a specific sample, i.e. y_i's are constants. We treat the sampling design/process as random, so s is a realization of a random variable S.

Sampling design tells us $\mathbb{P}(S=s)$ which captures the **inclusion probabilities**.

Definition 1.9 (**Inclusion Probabilities**).

$$\pi_i = \mathbb{P}(i \in S), \quad \pi_{ij} = \mathbb{P}(i,j\in S), \quad i \neq j \in U.$$

Then, we can define <u>estimators</u> as functions of S:

- $\tilde{\mu} = \frac{1}{n}\sum_{i\in S} y_i$ (sample mean estimator)
- $\tilde{\tau} = \sum_{i\in S} y_i$ (sample total estimator)
- $\tilde{\sigma}^2 = \frac{1}{n-1}\sum_{i\in S}(y_i-\tilde{\mu})^2$ (sample variance estimator)

Definition 1.10 (**Simple Random Sampling (SRS)**).

SRS is also known as **SRS without replacement (SRSWOR)**. Take a sample of size n out of N, such that all possible samples of size n are equally likely.

Note. The randomness in SRS comes from the sampling design, not the responses.

Example. Let $N = 4$, $n = 2$, then there are $\binom{4}{2} = 6$ possible samples:

$$\{1,2\} \quad \{1,3\} \quad \{1,4\} \quad \{2,3\} \quad \{2,4\} \quad \{3,4\},$$

each with probability $\frac{1}{6}$, e.g. $\mathbb{P}(S = \{1,2\}) = \frac{1}{6}$.

Question: How many possible samples of size n?

Answer: $\binom{N}{n} = \frac{N!}{n!(N-n)!}$.

Thus, we have

$$\mathbb{P}(S = s) = \frac{1}{\binom{N}{n}} \quad \text{for any subset } s \text{ of size } n.$$

We can also compute inclusion probabilities:

$$\pi_i = \mathbb{P}(i \in S) = \frac{\text{\# of samples with unit } i \text{ in it}}{\text{total \# of possibilities}} = \frac{\binom{1}{1}\binom{N-1}{n-1}}{\binom{N}{n}} = \frac{n}{N}.$$

Note. $\binom{1}{1}$ is the number of ways to choose unit i (to include i, we must pick i, so 1 way), and $\binom{N-1}{n-1}$ is the number of ways to choose the remaining $n-1$ units from the other $N-1$ units.

$$\pi_{ij} = \mathbb{P}(i, j \in S) = \frac{\text{\# of samples with units } i, j \text{ in it}}{\text{total \# of possibilities}} = \frac{\binom{2}{2}\binom{N-2}{n-2}}{\binom{N}{n}} = \frac{n(n-1)}{N(N-1)}.$$

Note. $\pi_{ij} \neq \pi_i \pi_j$ because sampling from a finite population without replacement means that the events $i \in S$ and $j \in S$ are not independent.

Lecture 3, 2025/09/10

R Intro & SRS in R

Here is some basic syntax for R:

```
x <- c()     # creates an empty vector
for (i in 1:50) {
x[i] <- 2*i      # fill it with some stuff
}
x   # look at what it now has
mean(x)
var(x)  # sample mean and variance
```

Obtaining SRS in R:

```
> N <- 10
> n <- 5
> x = c(1:10)
> sample(x,5, replace=F)     # replace=F indicates without replacement
[1] 6 2 4 7 8
> sample(x,5, replace=F)
[1] 8 4 5 6 3
```

Probability Review

Let X be a discrete random variable. Then,

(1) $\mathbb{E}[X] = \sum_x x\mathbb{P}(X = x), \quad \mathbb{E}[g(X)] = \sum_x g(x)\mathbb{P}(X = x)$.

(2) $\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] = \mathbb{E}\left[X^2\right] - (\mathbb{E}[X])^2$.

(3) Linearity: $\mathbb{E}\left[\sum_i^n a_i X_i\right] = \sum_i^n a_i \mathbb{E}[X_i]$.

(4) $\mathrm{Var}\left(\sum_i^n a_i X_i\right) = \sum_i^n a_i^2 \mathrm{Var}(X_i) + 2\sum_i^n \sum_{j\neq i} a_i a_j \mathrm{Cov}(X_i, X_j)$.

(5) $\mathrm{Cov}(X_i, X_j) = \mathbb{E}[X_i X_j] - \mathbb{E}[X_i]\,\mathbb{E}[X_j]$.

(6) If X_i and X_j are independent, then $\mathrm{Cov}(X_i, X_j) = 0$.

Example. Let's define an indicator random variable. Let

$$I_i = \begin{cases} 1, & \text{if unit } i \text{ is selected in the sample,} \\ 0, & \text{otherwise.} \end{cases}$$

Then, $\mathbb{P}(I_i = 1) = \mathbb{P}(i \in S) = \pi_i$ and $\mathbb{P}(I_i = 0) = 1 - \pi_i$. Thus,

$$\begin{aligned} \mathbb{E}\left[I_i\right] &= 0 \cdot (1 - \pi_i) + 1 \cdot \pi_i = \pi_i, \\ \mathbb{E}\left[I_i^2\right] &= 0^2 \cdot (1 - \pi_i) + 1^2 \cdot \pi_i = \pi_i, \\ \operatorname{Var}(I_i) &= \mathbb{E}\left[I_i^2\right] - (\mathbb{E}\left[I_i\right])^2 = \pi_i - \pi_i^2 = \pi_i(1 - \pi_i). \end{aligned}$$

For $i \neq j$, consider

$$I_i I_j = \begin{cases} 1, & \text{if both } i \text{ and } j \text{ are selected in the sample,} \\ 0, & \text{otherwise.} \end{cases}$$

Then, $\mathbb{P}(I_i I_j = 1) = \mathbb{P}(i, j \in S) = \pi_{ij}$ and $\mathbb{P}(I_i I_j = 0) = 1 - \pi_{ij}$. Thus,

$$\begin{aligned} \mathbb{E}\left[I_i I_j\right] &= 0 \cdot (1 - \pi_{ij}) + 1 \cdot \pi_{ij} = \pi_{ij}, \\ \operatorname{Cov}\left(I_i, I_j\right) &= \mathbb{E}\left[I_i I_j\right] - \mathbb{E}\left[I_i\right] \mathbb{E}\left[I_j\right] = \pi_{ij} - \pi_i \pi_j. \end{aligned}$$

So, we can write $\tilde{\mu}$ in terms of indicators:

$$\tilde{\mu} = \frac{1}{n} \sum_{i \in S} y_i = \frac{1}{n} \sum_{i \in U} I_i y_i = \frac{1}{n} \sum_{i=1}^{N} I_i y_i.$$

Definition 1.11 (**Bias, Unbiased Estimator**).

Suppose that $\tilde{\theta}$ is an estimator for θ. Then, the **bias** of $\tilde{\theta}$ is

$$\operatorname{Bias}\left(\tilde{\theta}\right) = \mathbb{E}\left[\tilde{\theta}\right] - \theta.$$

$\tilde{\theta}$ is an **unbiased estimator** for θ if $\operatorname{Bias}\left(\tilde{\theta}\right) = 0$, i.e. $\mathbb{E}\left[\tilde{\theta}\right] = \theta$.

Question: What are $\mathbb{E}[\tilde{\mu}]$ and $\operatorname{Var}(\tilde{\mu})$ for SRS?

Answer:

$$\begin{aligned}\mathbb{E}[\tilde{\mu}] &= \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{N} I_i y_i\right] = \frac{1}{n}\sum_{i=1}^{N} y_i \underbrace{\mathbb{E}[I_i]}_{\pi_i} \\ &= \frac{1}{n}\sum_{i=1}^{N} y_i \frac{n}{N} = \frac{1}{N}\sum_{i=1}^{N} y_i = \mu.\end{aligned}$$

So $\tilde{\mu}$ is an unbiased estimator for μ.

For SRS,

$$\operatorname{Cov}(I_i, I_j) = \pi_{ij} - \pi_i\pi_j = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n(N-n)}{N^2(N-1)}.$$

Then,

$$\begin{aligned}\operatorname{Var}(\tilde{\mu}) &= \operatorname{Var}\left(\frac{1}{n}\sum_{i=1}^{N} I_i y_i\right) \\ &= \frac{1}{n^2}\left[\sum_{i=1}^{N} y_i^2 \underbrace{\operatorname{Var}(I_i)}_{\pi_i(1-\pi_i)} + 2\sum_{i=1}^{N}\sum_{j\neq i} y_i y_j \operatorname{Cov}(I_i, I_j)\right] \\ &= \frac{1}{n^2}\left(1-\frac{n}{N}\right)\frac{n}{N}\left[\sum_{i=1}^{N} y_i^2 - \frac{1}{N-1}\sum_{i=1}^{N}\sum_{j\neq i} y_i y_j\right]\end{aligned}$$

Note that $\sum_{i=1}^{N}\sum_{j\neq i} y_i y_j = \sum_{i=1}^{N} y_i \sum_{j\neq i} y_j = \sum_{i=1}^{N} y_i(N\mu - y_i) = N\mu\sum_{i=1}^{N} y_i - \sum_{i=1}^{N} y_i^2 = N^2\mu^2 - \sum_{i=1}^{N} y_i^2.$

$$\begin{aligned} &= \frac{1}{n^2}\left(1-\frac{n}{N}\right)\frac{n}{N}\left[\sum_{i=1}^{N} y_i^2 - \frac{1}{N-1}\left(N^2\mu^2 - \sum_{i=1}^{N} y_i^2\right)\right] \\ &= \frac{1}{n^2}\left(1-\frac{n}{N}\right)\frac{n}{N}\frac{1}{N-1}\left[((N-1)+1)\sum_{i=1}^{N} y_i^2 - N^2\mu^2\right] \\ &= \frac{1}{n^2}\left(1-\frac{n}{N}\right)\frac{n}{N}N\frac{1}{N-1}\left(\sum_{i=1}^{N} y_i^2 - N\mu^2\right) \\ &= \left(1-\frac{n}{N}\right)\frac{\sigma^2}{n}.\end{aligned}$$

Thus for SRS, $\mathbb{E}[\tilde{\mu}] = \mu$ and $\operatorname{Var}(\tilde{\mu}) = \left(1-\frac{n}{N}\right)\frac{\sigma^2}{n}$, where $\left(1-\frac{n}{N}\right)$ is called the **finite population correction (fpc)**.

To get CIs, we estimate $\operatorname{Var}(\tilde{\mu})$ with $\widehat{\operatorname{Var}}(\tilde{\mu}) = \left(1-\frac{n}{N}\right)\frac{\hat{\sigma}^2}{n}$, as it turns out that $\mathbb{E}\left[\hat{\sigma}^2\right] = \sigma^2$.

Lecture 4, 2025/09/15

Exercise: Show that $\mathbb{E}\left[\hat{\sigma}^2\right] = \sigma^2$ (PS2).

In practice, we need to estimate $\operatorname{Var}(\tilde{\mu})$ because we don't know σ^2.

Question: What is an unbiased estimator of $\operatorname{Var}(\tilde{\mu})$?

Answer: Note that $\mathbb{E}\left[\left(1-\frac{n}{N}\right)\frac{\hat{\sigma}^2}{n}\right] = \left(1-\frac{n}{N}\right)\frac{\sigma^2}{n}$ (hint for A1 Q3).

Suppose we now have data from SRS, and we want an estimate of the population mean and 95% CI. We care about CIs because we want to know how uncertain our estimate is.

Definition 1.12 (**Standard Error (SE)**).
Standard error (SE) is an estimate of the standard deviation (of an estimator) based on a specific sample.

Example. For μ based on a SRS, the estimate is $\hat{\mu}$. We estimate $\operatorname{Var}(\tilde{\mu})$ with

$$\widehat{\operatorname{Var}}(\tilde{\mu}) = \left(1-\frac{n}{N}\right)\frac{\hat{\sigma}^2}{n},$$

and so $\operatorname{SE}(\hat{\mu}) = \sqrt{\widehat{\operatorname{Var}}(\tilde{\mu})} = \sqrt{\left(1-\frac{n}{N}\right)\frac{\hat{\sigma}^2}{n}}$, note that "hat" is used for σ^2.
For sampling, we will use Normal approximation to construct CIs, e.g. 95% CI for μ is

$$\hat{\mu} \pm 1.96\,\operatorname{SE}(\hat{\mu}).$$

Example. For τ (population total) in SRS, since $\tau = \sum_{i\in U} y_i = N\mu$, we have $\hat{\tau} = N\hat{\mu}$,

$$\operatorname{Var}(\tilde{\tau}) = \operatorname{Var}(N\tilde{\mu}) = N^2\operatorname{Var}(\tilde{\mu}) = N^2\left(1-\frac{n}{N}\right)\frac{\sigma^2}{n},$$

so $\widehat{\operatorname{Var}}(\tilde{\tau}) = N^2\widehat{\operatorname{Var}}(\tilde{\mu})$ and $\operatorname{SE}(\hat{\tau}) = N\cdot\operatorname{SE}(\hat{\mu})$. An approximate 95% CI for τ is

$$N\hat{\mu} \pm 1.96N\cdot\operatorname{SE}(\hat{\mu}).$$

Example (Special Case).

For proportion, suppose response is binary i.e.

$$y_i = \begin{cases} 1, & \text{answered yes,} \\ 0, & \text{answered no.} \end{cases}$$

The population mean of y_i's is called a **proportion**. Denote the population proportion as p. In this case,

$$\begin{aligned} \sigma^2 &= \frac{1}{N-1}\left(\sum_{i\in U} y_i^2 - N\mu^2\right) \\ &= \frac{1}{N-1}\left(Np - Np^2\right) \qquad (\text{since } y_i^2 = y_i \text{ and } \mu = p) \\ &= \frac{N}{N-1}p(1-p). \end{aligned}$$

Based on SRS,

$$\hat{\sigma}^2 = \frac{n}{n-1}\hat{p}(1-\hat{p}),$$

where $\hat{p}$ is the sample proportion. So a 95% CI for p is

$$\hat{p} \pm 1.96\,\mathrm{SE}\,(\hat{p}) = \hat{p} \pm 1.96\sqrt{\left(1-\frac{n}{N}\right)\frac{\hat{p}(1-\hat{p})}{n-1}}.$$

Note. $\frac{n}{N}$ is the **sampling fraction** and $1-\frac{n}{N}$ is the **fpc**. So if $n \approx N$, then $\mathrm{Var}\,(\tilde{\mu}) \to 0$, i.e. variability due to sampling $\to 0$. If $N \gg n$, then $\text{fpc} = 1 - \frac{n}{N} \approx 1$.

Recall: How many Americans ($N = 342M$) vs. Canadians ($N = 41.5M$) for a sample? If $\hat{\sigma}^2$ is similar for US/Canada, then $\mathrm{SE}\,(\hat{\mu})$ is also very similar, even though N is much larger for US.

Example (Sample Size Calculation).

Suppose we have $\hat{\sigma}^2$ under SRS, how wide is my 95% CI if

(a) Sampled 5% of population, i.e. $\frac{n}{N} = 0.05$?

(b) Sampled 100 people?

Proof.

(a) We don't know since we need to know sample size n or population size N.

(b) We don't know since we need to know $\frac{n}{N}$ or N.

$\square$

We need information on both n and N!

For the 95% CI for μ, the total CI width is

$$2 \cdot 1.96 \cdot \sqrt{\left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}^2}{n}}.$$

So if we want the total width $< c$, this means

$$\begin{aligned} \left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}^2}{n} &< \left(\frac{c}{2 \cdot 1.96}\right)^2 \\ \frac{1}{n} - \frac{1}{N} &< \frac{1}{\hat{\sigma}^2} \cdot \left(\frac{c}{2 \cdot 1.96}\right)^2 \\ n &> \left[\frac{1}{\hat{\sigma}^2}\left(\frac{c}{2 \cdot 1.96}\right) + \frac{1}{N}\right]^{-1}. \end{aligned}$$

Don't forget to round n to integer!

Where to get $\hat{\sigma}^2$?

- Previous survey.
- In the special case of proportions,

$$\hat{\sigma}^2 = \frac{n}{n-1}\hat{p}(1 - \hat{p}) \approx p(1 - p) \leq 0.5(1 - 0.5) = 0.25.$$

This means that we can use 0.25 as a conservative upper bound for $\hat{\sigma}^2$.

Analyzing a SRS in R

The sampling designs in this class can be analyzed using the R package survey.

```
install.packages("survey")  # only need to run once
library(survey) # load the package

srs <- svydesign(ids=~1, data = dat, fpc = ~N)
# "ids=~1" tells R its a SRS, no clusters or strata
# "fpc" tells R which column is pop size to use for fpc
svymean(~sleep, srs)
# get estimate and SE for mean of variable "sleep"
```

```
mean(dat$sleep)
sqrt((1 - 1249/48415) * var(dat$sleep)/1249)
# can also manually calculate mean and SE like this
confint(svymean(~sleep, srs))
# get 95% CI for mean of "sleep"
```

Lecture 5, 2025/09/17

1.3 Questionnaire Design

When a survey respondent reads or hears a survey question, the following process usually occurs in their mind:

- Comprehension (interpreting the question)
- Retrieval (recall of information)
- Judgment (summarizing the information)
- Reporting (answering the question in the required format)

A well-designed questionnaire should make these steps as easy and efficient as possible for the respondent. If the process breaks down anywhere, answers collected will likely be unreliable (measurement error).

Comprehension problems:

- Poor or ambiguous wording.
- Too complex.
- Leading questions.
- False assumptions or inferences.

Retrieval problems:

- Lack of retrieval cues.
- Things are too hard to remember.
- Events happened too long ago.

Judgment and reporting problems:

- Response categories are too specific or don't cover all possibilities.
- Difficult to estimate based on question constraints.
- Stigma of giving certain responses.
- Deliberate misreporting.
  - People don't want to answer sensitive questions or reluctant to admit.

Response scale effects:

- People often don't want to pick options on the ends.
- Scale of -5 to $+5$ is interpreted differently than 0 to 10. Respondents tend to use higher ratings on the -5 to $+5$ scale.

Types of responses (respondents may be asked to answer in the following ways):

- Open-ended, with numerical response.
- Closed questions with ordered response scales.
- Closed questions with categorical response options.
  - Should be disjoint, and cover all possibilities.
  - Be specific.
  - Order matters: some respondents stop reading when they see an option that seems to describe them.

Strategies for sensitive questions:

- Give them a reasonable excuse to soften the behavior.
- Make it as an anonymous as possible.
- Randomized response.

Question testing:

- Survey questions often need multiple rounds of testing and feedback.

1.4 Ratio and Regression Estimation

Suppose we measure 2 variables/attributes, and we have the following:

- Frame: $U = \{1, 2, \dots, N\}$.
- Values: $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)$.
- Population quantities:
  (1) $\mu_X = \frac{1}{N} \sum_{i \in U} x_i$.
  (2) $\mu_Y = \frac{1}{N} \sum_{i \in U} y_i$.
  (3) Population correlation coefficient: $\rho = \frac{\sum_{i \in U}(x_i - \mu_X)(y_i - \mu_Y)}{(N-1)\sigma_x \sigma_y}$.
  (4) Population ratio of means: $\theta \overset{\text{def.}}{\equiv} \frac{\mu_Y}{\mu_X} = \frac{\frac{1}{N}\sum_{i \in U} y_i}{\frac{1}{N}\sum_{i \in U} x_i} = \frac{\tau_y}{\tau_x}$.

From an SRS of size n, we can estimate the following quantities:

(1) Sample correlation:
$$(r =)\, \hat{\rho} = \frac{\sum_{i \in S}(x_i - \hat{\mu}_x)(y_i - \hat{\mu}_y)}{(n-1)\hat{\sigma}_x \hat{\sigma}_y}.$$

(2) Sample ratio of means:
$$\hat{\theta} = \frac{\hat{\mu}_y}{\hat{\mu}_x}.$$

Question: Why consider θ?

(1) We want to estimate $\frac{\mu_Y}{\mu_X}$.

> **Example.** Let
>
> $$y_i = \text{\# of gators in tract } i,$$
> $$x_i = \text{area of tract } i \text{ (in acres)}.$$
>
> Then, θ is the average # of alligators per acre.

Example. Let

$$y_i = \text{\# served by employee } i \text{ in their shift,}$$
$$x_i = \text{length of that shift (in hours).}$$

Then, θ is the average # served per hour.

(2) When we know μ_X (population mean in this case), and we want to use it to get a more accurate estimate of μ_Y.

- If we don't know μ_X, we just estimate μ_Y with $\hat{\mu}_Y$.
- If we know μ_X, we can estimate μ_Y with $\hat{\mu}_Y \cdot \frac{\mu_X}{\hat{\mu}_X}$, in which we can think of $\frac{\mu_X}{\hat{\mu}_X}$ as an adjustment factor. Also,

$$\hat{\mu}_Y \cdot \frac{\mu_X}{\hat{\mu}_X} = \hat{\theta} \cdot \mu_X$$

  which works well if x and y are highly correlated.

Properties of Estimator $\tilde{\theta} = \frac{\tilde{\mu}_Y}{\tilde{\mu}_X}$

Goal: Find $\mathbb{E}\left[\tilde{\theta}\right]$ and $\mathrm{Var}\left(\tilde{\theta}\right)$ under SRS. This is not trivial since $\mathbb{E}\left[\frac{W}{V}\right] \neq \frac{\mathbb{E}[W]}{\mathbb{E}[V]}$ in general.

Let's express $\tilde{\mu}_Y$ as the following:

$$\underbrace{\tilde{\mu}_Y}_{\text{r.v.}} = \underbrace{\mu_Y}_{\text{constant}} (1 + \underbrace{\varepsilon_Y}_{\text{r.v.}}).$$

Then,

$$\mu_Y = \mathbb{E}\left[\tilde{\mu}_Y\right] = \mu_Y(1 + \mathbb{E}\left[\varepsilon_Y\right]) \implies \mathbb{E}\left[\varepsilon_Y\right] = 0.$$

Similarly, we can express $\tilde{\mu}_X$ as

$$\tilde{\mu}_X = \mu_X(1 + \varepsilon_X), \quad \text{so } \mathbb{E}\left[\varepsilon_X\right] = 0.$$

We can write

$$\begin{aligned}
\tilde{\theta} = \frac{\tilde{\mu}_Y}{\tilde{\mu}_X} &= \frac{\mu_Y(1+\varepsilon_Y)}{\mu_X(1+\varepsilon_X)} \\
&= \frac{\mu_Y}{\mu_X}\cdot(1+\varepsilon_Y)(1-\varepsilon_X+\varepsilon_X^2-\varepsilon_X^3+\dots) && \text{by geometric series} \\
&= \frac{\mu_Y}{\mu_X}\left(1-\varepsilon_X+\varepsilon_Y-\varepsilon_X\varepsilon_Y+\varepsilon_X^2+\varepsilon_Y\varepsilon_X^2-\cdots\right) \\
&\approx \frac{\mu_Y}{\mu_X}(1-\varepsilon_X+\varepsilon_Y) && \text{by ignoring higher order terms.}
\end{aligned}$$

So, we have

$$\mathbb{E}\left[\tilde{\theta}\right] \approx \frac{\mu_Y}{\mu_X}(1-\mathbb{E}\left[\varepsilon_X\right]+\mathbb{E}\left[\varepsilon_Y\right]) = \frac{\mu_Y}{\mu_X}.$$

Thus, $\tilde{\theta}$ is approximately unbiased for θ. For the variance, we have

$$\begin{aligned}
\operatorname{Var}\left(\tilde{\theta}\right) &\approx \left(\frac{\mu_Y}{\mu_X}\right)^2 \operatorname{Var}\left(1-\varepsilon_X+\varepsilon_Y\right) \\
&= \left(\frac{\mu_Y}{\mu_X}\right)^2 \operatorname{Var}\left(1-\left(\frac{\tilde{\mu}_X}{\mu_X}-1\right)+\left(\frac{\tilde{\mu}_Y}{\mu_Y}-1\right)\right) \\
&= \left(\frac{\mu_Y}{\mu_X}\right)^2\cdot\frac{1}{\mu_Y^2}\operatorname{Var}\left(\tilde{\mu}_Y-\mu_Y\cdot\frac{\tilde{\mu}_X}{\mu_X}\right) \\
&= \frac{1}{\mu_X^2}\operatorname{Var}\left(\tilde{\mu}_Y-\theta\tilde{\mu}_X\right).
\end{aligned}$$

Note that

$$\tilde{\mu}_Y-\theta\tilde{\mu}_X = \frac{1}{n}\sum_{i\in S} y_i - \theta\cdot\frac{1}{n}\sum_{i\in S} x_i = \frac{1}{n}\sum_{i\in S}(y_i-\theta x_i) = \frac{1}{n}\sum_{i\in S} r_i = \tilde{\mu}_r.$$

Hence, we know

$$\operatorname{Var}\left(\tilde{\mu}_r\right) = \left(1-\frac{n}{N}\right)\frac{\sigma_r^2}{n}, \quad \text{from SRS theory.}$$

So,

$$\operatorname{Var}\left(\tilde{\theta}\right) \approx \frac{1}{\mu_X^2}\left(1-\frac{n}{N}\right)\frac{\sigma_r^2}{n}$$

where

$$
\begin{aligned}
\sigma_r^2 &= \frac{1}{N-1}\sum_{i\in U}(r_i-\mu_r)^2 \\
&= \frac{1}{N-1}\sum_{i\in U}\left[(y_i-\theta x_i)-(\mu_Y-\theta\mu_X)\right]^2 \\
&= \frac{1}{N-1}\sum_{i\in U}\left[(y_i-\mu_Y)-\theta(x_i-\mu_X)\right]^2 \\
&= \frac{1}{N-1}\sum_{i\in U}\left[(y_i-\mu_Y)^2+\theta^2(x_i-\mu_X)^2-2\theta(y_i-\mu_Y)(x_i-\mu_X)\right] \\
&= \sigma_Y^2+\theta^2\sigma_X^2-2\theta\rho\sigma_X\sigma_Y.
\end{aligned}
$$

Therefore, we have

$$
\begin{aligned}
\mathbb{E}\left[\tilde{\theta}\right] &\approx \theta \\
\operatorname{Var}\left(\tilde{\theta}\right) &\approx \frac{1}{\mu_X^2}\left(1-\frac{n}{N}\right)\frac{1}{n}\left(\sigma_Y^2+\theta^2\sigma_X^2-2\theta\rho\sigma_X\sigma_Y\right).
\end{aligned}
$$

So to estimate θ based on SRS, we use

$$
\hat{\theta}=\frac{\hat{\mu}_Y}{\hat{\mu}_X}, \qquad \operatorname{SE}\left(\hat{\theta}\right)=\sqrt{\frac{1}{\hat{\mu}_X^2}\left(1-\frac{n}{N}\right)\frac{1}{n}\left(\hat{\sigma}_Y^2+\hat{\theta}^2\hat{\sigma}_X^2-2\hat{\theta}\hat{\rho}\hat{\sigma}_X\hat{\sigma}_Y\right)}.
$$

And a 95% CI for θ is approximately

$$
\hat{\theta}\pm 1.96\operatorname{SE}\left(\hat{\theta}\right).
$$

Lecture 7, 2025/09/24

We can estimate $\operatorname{Var}\left(\tilde{\theta}\right)$ using

$$
\widehat{\operatorname{Var}}(\tilde{\theta})=\frac{1}{\hat{\mu}_X^2}\left(1-\frac{n}{N}\right)\frac{1}{n}\left(\hat{\sigma}_Y^2+\hat{\theta}^2\hat{\sigma}_X^2-2\hat{\theta}\hat{\rho}\hat{\sigma}_X\hat{\sigma}_Y\right).
$$

Question: When will the estimation work well, i.e. when will $\operatorname{Var}\left(\tilde{\theta}\right)$ be small?

Answer:

- We want ρ to be large, i.e. $\rho \approx 1$.

Also note that

$$\sigma_Y^2 + \theta^2\sigma_X^2 - 2\theta\rho\sigma_X\sigma_Y = \sigma_r^2 = \frac{1}{N-1}\sum_{i\in U}(y_i - \theta x_i - (\mu_Y - \underbrace{\theta\mu_X}_{\mu_Y}))^2 = \frac{1}{N-1}\sum_{i\in U}(y_i - \theta x_i)^2.$$

- We want $y_i \approx \theta x_i$ for each unit i, i.e. y values are proportional to x values (can be described well as a straight line through the origin with slope θ).

Recall from last time that the CI for θ is $\hat{\theta} \pm 1.96\,\mathrm{SE}\left(\hat{\theta}\right)$. If we are interested in θ itself, we may also use $\tilde{\theta}$ to improve the accuracy of the estimate of μ_Y when μ_X is known. Since $\mu_Y = \theta\mu_X$, we can use $\tilde{\theta}\mu_X$ as an estimator for μ_Y. Then,

$$\begin{aligned}
\mathbb{E}\left[\tilde{\theta}\mu_X\right] &= \mu_X\mathbb{E}\left[\tilde{\theta}\right] \approx \mu_X\frac{\mu_Y}{\mu_X} = \mu_Y,\\
\mathrm{Var}\left(\tilde{\theta}\mu_X\right) &= \mu_X^2\,\mathrm{Var}\left(\tilde{\theta}\right).
\end{aligned}$$

Thus, we can construct the ratio estimate of μ_Y (if μ_X is known) as

$$\hat{\mu}_{\text{ratio}} = \hat{\theta}\mu_X = \frac{\hat{\mu}_Y}{\hat{\mu}_X}\mu_X.$$

The estimate of the variance is

$$\widehat{\mathrm{Var}}(\tilde{\mu}_{\text{ratio}}) = \mu_X^2\widehat{\mathrm{Var}}(\tilde{\theta}) = \frac{\mu_X^2}{\hat{\mu}_X^2}\left(1-\frac{n}{N}\right)\frac{1}{n}\left(\hat{\sigma}_Y^2 + \hat{\theta}^2\hat{\sigma}_X^2 - 2\hat{\theta}\hat{\rho}\hat{\sigma}_X\hat{\sigma}_Y\right) \overset{\text{OR}}{\approx} \left(1-\frac{n}{N}\right)\frac{1}{n}\left(\hat{\sigma}_Y^2 + \hat{\theta}^2\hat{\sigma}_X^2 - 2\hat{\theta}\hat{\rho}\hat{\sigma}_X\hat{\sigma}_Y\right).$$

Regression Estimation of the Mean

Consider the SRS with (x_i, y_i) measured and μ_X known, and the estimator

$$\tilde{\mu}_* = \tilde{\mu}_Y + b(\mu_X - \tilde{\mu}_X)$$

where b is a constant. Note that

- when $b = 0$, $\tilde{\mu}_* = \tilde{\mu}_Y$, the sample mean estimator for μ_Y (does not use any info about x).
- when $b = \tilde{\theta} = \frac{\tilde{\mu}_Y}{\tilde{\mu}_X}$, $\tilde{\mu}_* = \tilde{\mu}_Y + \frac{\tilde{\mu}_Y}{\tilde{\mu}_X}\mu_X - \tilde{\mu}_Y = \tilde{\theta}\mu_X = \hat{\mu}_{\text{ratio}}$.

And for any value of b, we have

$$\begin{aligned}\mathbb{E}[\tilde{\mu}_*] &= \mathbb{E}[\tilde{\mu}_Y] + b(\mu_X - \mathbb{E}[\tilde{\mu}_X]) \\ &= \mu_Y + b(\mu_X - \mu_X) \\ &= \mu_Y, \text{ i.e. unbiased for any } b.\end{aligned}$$

So it makes sense to choose b to minimize $\mathrm{Var}(\tilde{\mu}_*)$. We have

$$\begin{aligned}\mathrm{Var}(\tilde{\mu}_*) &= \mathrm{Var}(\tilde{\mu}_Y + b(\mu_X - \tilde{\mu}_X)) \\ &= \mathrm{Var}(\tilde{\mu}_Y - b\tilde{\mu}_X).\end{aligned}$$

Note that this is the same form as for the ratio estimator except with $b = \theta$. So, we use the same approach to get

$$\mathrm{Var}(\tilde{\mu}_*) = \left(1 - \frac{n}{N}\right)\frac{1}{n}\left(\sigma_Y^2 + b^2\sigma_X^2 - 2b\rho\sigma_X\sigma_Y\right).$$

Let's find the optimal b to minimize this <u>estimated</u> variance, let

$$f(b) = \hat{\sigma}_Y^2 + b^2\hat{\sigma}_X^2 - 2b\hat{\rho}\hat{\sigma}_X\hat{\sigma}_Y$$

based on a sample. Then,

$$f'(b) = 2b\hat{\sigma}_X^2 - 2\hat{\rho}\hat{\sigma}_X\hat{\sigma}_Y \overset{\text{set}}{=} 0 \implies b = \frac{\hat{\rho}\hat{\sigma}_Y}{\hat{\sigma}_X}.$$

We can check this indeed minimizes $f(b)$ since $f''(b) = 2\hat{\sigma}_X^2 > 0$.

Let's call

$$\hat{\mu}_{\text{reg}} = \hat{\mu}_Y + \frac{\hat{\rho}\hat{\sigma}_Y}{\hat{\sigma}_X}(\mu_X - \hat{\mu}_X)$$

the <u>regression estimate of μ_Y</u>. The estimate of the variance is

$$\widehat{\mathrm{Var}}(\tilde{\mu}_{\text{reg}}) = \left(1 - \frac{n}{N}\right)\frac{1}{n}\left(\hat{\sigma}_Y^2 + \frac{\hat{\rho}^2\hat{\sigma}_Y^2}{\hat{\sigma}_X^2}\hat{\sigma}_X^2 - 2\hat{\rho}\frac{\hat{\rho}\hat{\sigma}_Y}{\hat{\sigma}_X}\hat{\sigma}_X\hat{\sigma}_Y\right) = \left(1 - \frac{n}{N}\right)\frac{1}{n}(1 - \hat{\rho}^2)\hat{\sigma}_Y^2.$$

Remark (**Three ways to estimate** μ_Y).

$\hat{\mu}_Y$, $\hat{\mu}_{\text{ratio}}$, and $\hat{\mu}_{\text{reg}}$.

Question 1: When is $\hat{\mu}_{\text{ratio}}$ better than $\hat{\mu}_Y$? (i.e. lower estimated variance)

Answer: Recall that

$$\begin{aligned}\widehat{\mathrm{Var}}(\tilde{\mu}_{\text{ratio}}) &\approx \left(1-\frac{n}{N}\right)\frac{1}{n}\left(\hat{\sigma}_Y^2+\hat{\theta}^2\hat{\sigma}_X^2-2\hat{\theta}\hat{\rho}\hat{\sigma}_X\hat{\sigma}_Y\right),\\ \widehat{\mathrm{Var}}(\hat{\mu}_Y) &= \left(1-\frac{n}{N}\right)\frac{\hat{\sigma}_Y^2}{n}.\end{aligned}$$

We want $\widehat{\mathrm{Var}}(\tilde{\mu}_{\text{ratio}}) < \widehat{\mathrm{Var}}(\tilde{\mu}_Y)$, i.e.

$$\begin{aligned}\hat{\sigma}_Y^2+\hat{\theta}^2\hat{\sigma}_X^2-2\hat{\theta}\hat{\rho}\hat{\sigma}_X\hat{\sigma}_Y &< \hat{\sigma}_Y^2\\ \hat{\theta}^2\hat{\sigma}_X^2-2\hat{\theta}\hat{\rho}\hat{\sigma}_X\hat{\sigma}_Y &< 0\\ \hat{\rho} > \frac{\hat{\theta}\hat{\sigma}_X}{2\hat{\sigma}_Y} &= \frac{1}{2}\frac{\hat{\mu}_Y\hat{\sigma}_X}{\hat{\mu}_X\hat{\sigma}_Y}.\end{aligned}$$

So if x, y have similar means/variances, then $\hat{\rho} > \frac{1}{2}$ is needed for ratio estimation to be better.

Question 2: When is $\hat{\mu}_{\text{reg}}$ better than $\hat{\mu}_Y$?

Answer: Recall that $\widehat{\mathrm{Var}}(\tilde{\mu}_{\text{reg}}) = \left(1-\frac{n}{N}\right)\frac{1}{n}(1-\hat{\rho}^2)\hat{\sigma}_Y^2$. We want $\widehat{\mathrm{Var}}(\tilde{\mu}_{\text{reg}}) < \widehat{\mathrm{Var}}(\tilde{\mu}_Y)$, i.e.

$$1-\hat{\rho}^2 < 1 \implies \hat{\rho}^2 > 0 \implies \hat{\rho} \neq 0.$$

As long as $\hat{\rho} \neq 0$, regression estimation is essentially always better than the sample mean.

Question 3: When is $\hat{\mu}_{\text{reg}}$ better than $\hat{\mu}_{\text{ratio}}$?

Answer: See A2 Q3.

Lecture 8, 2025/09/29

1.5 Stratified Sampling

Divide the population of size N into H strata (subpopulations) such that

$$N = \sum_{h=1}^{H} N_h$$

where N_h is the population size of stratum h. We have

Stratum	Units	Measurements/Values
1	$\{1, 2, \dots, N_1\}$	$\{y_{11}, y_{12}, \dots, y_{1N_1}\}$
2	$\{1, 2, \dots, N_2\}$	$\{y_{21}, y_{22}, \dots, y_{2N_2}\}$
$\vdots$	$\vdots$	$\vdots$
H	$\{1, 2, \dots, N_H\}$	$\{y_{H1}, y_{H2}, \dots, y_{HN_H}\}$

Remark. Stratified sampling works well when units in a stratum are similar among themselves.

Assume we take a SRS of size n_h from N_h in each stratum $h = 1, 2, \dots, H$ independently.

Population quantities:

- μ: overall population mean (over all N units).
- μ_h: population mean of stratum h.
- σ_h^2: population variance of stratum h.

Sample quantities:

- $\hat{\mu}_h$: sample mean of stratum h.
- $\hat{\sigma}_h^2$: sample variance of stratum h.

The key quantity to estimate is μ. We can write μ as:

$$\mu = \frac{1}{N} \sum_{h=1}^{H} \sum_{j=1}^{N_h} y_{hj} = \frac{1}{N} \sum_{h=1}^{H} N_h \mu_h = \sum_{h=1}^{H} \frac{N_h}{N} \mu_h.$$

where $\sum_{j=1}^{N_h} y_{hj}$ is the population total of stratum h, $\frac{N_h}{N}$ is the fraction of population that belongs to stratum h. Let $W_h = \frac{N_h}{N}$ be the “stratum weights”. Then,

$$\mu = \sum_{h=1}^{H} W_h \mu_h.$$

So a natural estimate for μ is

$$\hat{\mu}_S = \sum_{h=1}^{H} W_h \hat{\mu}_h.$$

Question: How many possible samples are there for stratified sampling vs. overall SRS?

Example. ($N = 80, n = 20$) vs. ($N_1 = N_2 = N_3 = N_4 = 20$ and $n_1 = n_2 = n_3 = n_4 = 5$).

- Overall SRS: $\binom{80}{20}$.
- Stratified SRS: $\binom{20}{5}^4$ where $\binom{20}{5}$ is the # of ways to choose 5 from 20 in each stratum.

Note that $\binom{80}{20} > \binom{20}{5}^4$. By stratifying, we ensure that we get exactly 5 from each stratum achieving a balanced sample which is not guaranteed with an overall SRS.

Mean and Variance of $\tilde{\mu}_S$

Note that

$$\mathbb{E}\left[\tilde{\mu}_S\right] = \mathbb{E}\left[\sum_{h=1}^{H} W_h \tilde{\mu}_h\right] = \sum_{h=1}^{H} W_h \mathbb{E}\left[\tilde{\mu}_h\right] = \sum_{h=1}^{H} W_h \mu_h = \mu$$

since we have SRS in each stratum. Thus, $\tilde{\mu}_S$ is unbiased for μ.

Also,

$$\begin{aligned}
\mathrm{Var}\left(\tilde{\mu}_S\right) = \mathrm{Var}\left(\sum_{h=1}^{H} W_h \tilde{\mu}_h\right) &= \sum_{h=1}^{H} W_h^2 \mathrm{Var}\left(\tilde{\mu}_h\right) \quad \text{(independent strata)} \\
&= \sum_{h=1}^{H} W_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{\sigma_h^2}{n_h}.
\end{aligned}$$

Allocation

For a fixed $n = \sum_{h=1}^{H} n_h$ (total sample size), how to choose $n_1, n_2, \ldots, n_H$?

Approach 1: Use the same sampling fraction in each stratum. This is called proportional allocation:

$$\frac{n}{N} = \frac{n_h}{N_h} \quad \forall h,$$

where $\frac{n}{N}$ is the overall sampling fraction and $\frac{n_h}{N_h}$ is the sampling fraction in stratum h.

Example. Let $N_1 = 100, N_2 = 200$, say we can sample a total of $n = 30$. Then,

$$\frac{n}{N} = \frac{30}{300} = 0.1 \implies n_1 = 10, n_2 = 20.$$

That is, $n_h = \frac{n}{N} N_h = W_h n \propto W_h$.

Under proportional allocation, we have

$$\begin{aligned}
\operatorname{Var}\left(\tilde{\mu}_{S,\mathrm{prop}}\right) &= \sum_{h=1}^{H} W_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{\sigma_h^2}{n_h} \\
&= \sum_{h=1}^{H} W_h \frac{N_h}{N} \frac{1}{n_h} \left(1 - \frac{n}{N}\right) \sigma_h^2 \\
&= \sum_{h=1}^{H} W_h \left(1 - \frac{n}{N}\right) \frac{\sigma_h^2}{n}.
\end{aligned}$$

Question: How does this compare to $\operatorname{Var}(\tilde{\mu})$ for an overall SRS (size n from N)?

Note that

$$\frac{\operatorname{Var}\left(\tilde{\mu}_{S,\mathrm{prop}}\right)}{\operatorname{Var}(\tilde{\mu})} = \frac{\sum_{h=1}^{H} W_h \left(1 - \frac{n}{N}\right) \frac{\sigma_h^2}{n}}{\left(1 - \frac{n}{N}\right) \frac{\sigma^2}{n}} = \frac{\sum_{h=1}^{H} W_h \sigma_h^2}{\sigma^2}.$$

Since $\sum_{h=1}^{H} W_h = 1$, stratified sampling is good if σ_h^2 (variance within strata) is small relative to σ^2 (overall population variance).

Lecture 9, 2025/10/01

Decompose the population variance σ^2 in stratified sampling:

$$\begin{aligned}
\sigma^2 &= \frac{1}{N-1} \sum_{h=1}^{H} \sum_{j=1}^{N_h} (y_{hj} - \mu)^2 \qquad \text{definition of pop. variance in stratified sampling} \\
&= \frac{1}{N-1} \sum_{h=1}^{H} \sum_{j=1}^{N_h} \left[(y_{hj} - \mu_h) + (\mu_h - \mu)\right]^2 \\
&= \frac{1}{N-1} \sum_{h=1}^{H} \sum_{j=1}^{N_h} \left[(y_{hj} - \mu_h)^2 + (\mu_h - \mu)^2 + 2(y_{hj} - \mu_h)(\mu_h - \mu)\right] \\
&= \frac{1}{N-1} \left[\sum_{h=1}^{H} \sigma_h^2 (N_h - 1) + \sum_{h=1}^{H} N_h (\mu_h - \mu)^2 + 2 \sum_{h=1}^{H} \left((\mu_h - \mu) \underbrace{\sum_{j=1}^{N_h} (y_{hj} - \mu_h)}_{= N_h \mu_h - N_h \mu_h = 0}\right)\right] \\
&= \sum_{j=1}^{N_h} \frac{N_h - 1}{N-1} \sigma_h^2 + \sum_{h=1}^{H} \frac{N_h}{N-1} (\mu_h - \mu)^2.
\end{aligned}$$

So if N_h and N are large, we have $\frac{N_h-1}{N-1} \approx \frac{N_h}{N} = W_h$. Thus,

$$\sigma^2 \approx \sum_{h=1}^{H} W_h \sigma_h^2 + \sum_{h=1}^{H} W_h (\mu_h - \mu)^2.$$

Note. $\sigma_h^2 = \dfrac{1}{N_h - 1} \displaystyle\sum_{j=1}^{N_h} (y_{hj} - \mu_h)^2$ is the population variance of stratum h and $N_h \mu_h$ is the population total of stratum h.

Recall that allocation is choosing $n_1, n_2, \dots, n_H$ for a fixed $n = \sum_{h=1}^{H} n_h$. The first approach is proportional allocation: $n_h \propto W_h$ and last time we found that

$$\begin{aligned} \frac{\operatorname{Var}\left(\tilde{\mu}_{S,\text{prop}}\right)}{\operatorname{Var}(\tilde{\mu})} = \frac{\sum_{h=1}^{H} W_h \sigma_h^2}{\sigma^2} &\approx \frac{\sum_{h=1}^{H} W_h \sigma_h^2}{\sum_{h=1}^{H} W_h \sigma_h^2 + \sum_{h=1}^{H} W_h (\mu_h - \mu)^2} \geq 0 \\ &\leq 1. \end{aligned}$$

Last time: Stratum works well if σ_h^2 are small relative to σ^2, turns out it's equivalent conceptually to if μ_h are very different compared to μ.

Approach 2: Optimal allocation (i.e. minimize variance) to choose $n_1, n_2, \dots, n_H$.

If we have some knowledge of σ_h^2, we can use that to set $n_1, \dots, n_H$ to minimize $\operatorname{Var}(\tilde{\mu}_S)$ for a fixed $n = \sum_{h=1}^{H} n_h$.

Problem: Minimize $\operatorname{Var}(\tilde{\mu}_S) = \displaystyle\sum_{h=1}^{H} W_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{\sigma_h^2}{n_h}$ as a function of $n_1, n_2, \dots, n_H$ with constraint $\displaystyle\sum_{h=1}^{H} n_h = n$. This can be solved with Lagrange multipliers. Setup:

$$L(n_1, n_2, \dots, n_H) = \sum_{h=1}^{H} W_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{\sigma_h^2}{n_h} + \lambda \left(\underbrace{\sum_{h=1}^{H} n_h - n}_{\text{constraint}=0} \right).$$

Set $\frac{\partial L}{\partial n_h} = 0$ for each $h = 1, \dots, H$, and $\frac{\partial L}{\partial \lambda} = 0$ to get (PS4 Q2):

$$n_h = \frac{W_h \sigma_h}{\sum_{h=1}^{H} W_h \sigma_h} \cdot n \propto W_h \sigma_h$$

and the denominator $\sum_{h=1}^{H} W_h \sigma_h$ does not depend on h.

Optimal allocation says we sample more from stratum with:

(1) Large N_h (large W_h).

(2) Large variance (σ_h high).

Example.

Stratum	W_h	$\hat{\sigma}_h$	$W_h\hat{\sigma}_h$
1	0.25	4	1
2	0.25	8	2
3	0.5	4	2

Note that we plug in our guess/estimate $\hat{\sigma}_h$ for σ_h in practice.

Say we can do $n = 100$ surveys in total.

(1) Proportional allocation: $n_1 = 25, n_2 = 25, n_3 = 50$.

(2) Optimal allocation: $n_1 = \frac{1}{5} \cdot 100 = 20, n_2 = \frac{2}{5} \cdot 100 = 40, n_3 = \frac{2}{5} \cdot 100 = 40$. And

$$\sum_{h=1}^{H} W_h\hat{\sigma}_h = \sum_{h=1}^{3} W_h\hat{\sigma}_h = 1 + 2 + 2 = 5.$$

Question: When is proportional allocation also optimal?

Answer: When σ_h are all the same, i.e. $\sigma_1 = \sigma_2 = \ldots = \sigma_H$.

Analyzing a Stratified Sample in R

To estimate the number of births in a country in a year, a researcher compiled a list of 158 hospitals in the country.

- The researcher divided hospitals into strata based on their size: 42 small, 99 medium, and 17 large.
- The researcher visited an SRS of 4 small, 5 medium, and 6 large hospitals, all sampled independently from their strata.

```
library(survey)
dat <- read.csv("births.csv")
boxplot(dat$births~dat$strata) # boxplots by strata
```

```
### Set up svydesign:
strat <- svydesign(id=~1, fpc=~N, strata=~strata, data=dat)
# the "strata" tells R where the strata are coded
# what's different about how "N" is specified?
svymean(~births, strat)
svytotal(~births, strat)
confint(svytotal(~births, strat))
# Can also look at totals by stratum:
svyby(~births, ~strata, strat, svytotal)
```

Stratified Sampling in Practice

To set up a stratified sampling design, we need:

- # of units in each stratum: $N_1, N_2, \ldots, N_H$.
- A list of units for each stratum.
- Each unit should belongg to exactly one stratum.
- (Optional) For optimal allocation, good guesses of σ_h^2; otherwise, use proportional allocation.

Why stratify?

- If it makes sense to divide the population into strata that are internally similar.
- To achieve desired representativeness from each stratum.
- Can get lower variance estimate using the same total sample size (compared to SRS on entire population), if the stratum means μ_h are very different from the overall mean μ. Equivalently, this means σ_h^2 within strata are smaller than the overall variance σ^2.

Why not stratify?

- It's too much work.

What if we are not sure how different the strata will be?

- Doesn't matter, if strata are large and n_h are reasonably large ($\geq$ 30 or so), then FPCs $\approx$ 1.
- Use proportional allocation.
- Under these conditions, stratified variance will not be worse than taking an overall SRS.

Poststratification

Suppose we know

- # of units in each stratum: $N_1, N_2, \dots, N_H$.
- Each unit belongs to exactly one stratum.

However, if our frame does not tell us which units belong to which stratum, we cannot take a stratified sample. But we can analyze the data as if it came from a stratified sampling design. This is called **poststratification**.

Steps to do poststratification:

(1) Decide strata beforehand just like for stratified sampling.

(2) Take a SRS from the entire population, and ask each sampled unit which stratum it belongs to.

(3) Use survey data to determine $n_1, n_2, \dots, n_H$.

(4) Analyze the data as if it came from a stratified sampling design, the post-stratified estimate of μ is

$$\hat{\mu}_{\text{post}} = \sum_{h=1}^{H} W_h \hat{\mu}_h.$$

(5) The poststratified variance can be approximated using the stratified variance formula for proportional allocation:

$$\widehat{\text{Var}}(\tilde{\mu}_{\text{post}}) = \sum_{h=1}^{H} W_h \left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}_h^2}{n}.$$

Example (**Poststratification Example**).

Suppose you take a survey of Vancouver, want to stratify by ethnicity. Instead, take SRS of $n = 1000$ since we don't have a list of ethnicities.

Stratum	W_h	n_h
White	0.48	$550 \to \hat{\mu}_1$
Chinese	0.27	$200 \to \hat{\mu}_2$
Other Asian	0.19	$150 \to \hat{\mu}_3$
Other	0.06	$100 \to \hat{\mu}_4$

Note that w_h are from the census and n_h represents the number after the survey. Then,

$$\hat{\mu}_{\text{post}} = 0.48\hat{\mu}_1 + 0.27\hat{\mu}_2 + 0.19\hat{\mu}_3 + 0.06\hat{\mu}_4.$$

μ_{post} adjusts estimate to each stratum's actual share of population.

Lecture 10, 2025/10/06

1.6 Cluster Sampling

Divide the population into clusters.

Sampling design: we sample clusters, then take measurements on some or all units in the sampled clusters.

Terms:

- PSU: primary sampling unit (i.e. cluster).
- SSU: secondary sampling unit (i.e. unit within cluster).

Example.

(1) Imagine we are doing a neighbourhood survey. We may have the following clusters.
  - PSU: block of houses or apartment buildings.
  - SSU: one residental unit.

(2) Cows on a farm. We may have:
  - PSU: farm.
  - SSU: cow.

Question: Why do we use cluster sampling?

Answer:

- For convenience, e.g. main cost is to access clusters.
- It is easy to sample/measure units in the same cluster.

In practice, units within the same cluster tend to be more similar among themselves compared to the population as a whole.

Notation: Assume the population is made up of A clusters, each with B units (equal-sized clusters), so the population size is $N = AB$.

First, take SRS of size a from A clusters, then take SRS of size b from B units in each sampled cluster. So, the total sample size is $n = ab$.

Case 1: $b = B$, i.e. measure all units of the sampled clusters.

Suppose we have $\mu_1, \mu_2, \dots, \mu_A$ are cluster means in the population. Then, the population mean is

$$\mu = \frac{1}{A}\sum_{a=1}^{A}\mu_a \qquad \text{since equal-sized clusters assumed.}$$

After sampling, we observe values μ_i where indices i are a SRS of size a from $\{1, 2, \dots, A\}$. From which, we estimate μ using

$$\hat{\mu}_{\text{clu}} = \frac{1}{a}\sum_{i\in S}\hat{\mu}_i, \quad \text{with } \operatorname{Var}(\tilde{\mu}_{\text{clu}}) = \left(1 - \frac{a}{A}\right)\frac{\sigma_A^2}{a}$$

where $\sigma_A^2 = \frac{1}{A-1}\sum_{a=1}^{A}(\mu_a - \mu)^2$ is the population variance of cluster means.

Example. There are 20 cows on 4 farms. We have the following data on weights of cows (in 1000's of lbs):

Farm	Individual weights					μ_a
1	1.3	1.3	1.6	1.4	1.4	1.4
2	1.2	1.4	1.3	1.1	1.5	1.3
3	1.4	1.6	1.5	1.5	1.5	1.5
4	1.0	1.1	1.3	0.9	1.2	1.1

Assume that $N = 20$ is the whole population. Then, the population mean of 20 cows is

$$\mu = \frac{1.4 + 1.3 + 1.5 + 1.1}{4} = 1.325$$

The population variance of 20 cows is

$$\sigma^2 = \frac{1}{20-1}\left[(1.3 - 1.325)^2 + (1.3 - 1.325)^2 + \cdots + (1.2 - 1.325)^2\right] = 0.03776.$$

The population variance of 4 farms means is

$$\sigma_A^2 = \frac{1}{4-1}\left[(1.4-1.325)^2 + (1.3-1.325)^2 + (1.5-1.325)^2 + (1.1-1.325)^2\right] = 0.02917.$$

Consider 2 sampling designs:

(1) SRS of 10 cows from 20 cows.

(2) SRS of 2 farms from 4 farms, weigh all cows in the sampled farms.

How do their estimation variances compare?

(1) $\left(1-\frac{10}{20}\right)\frac{0.03776}{10} = 0.0019.$

(2) $\left(1-\frac{2}{4}\right)\frac{0.02917}{2} = 0.0073.$

Note that the second variance is almost 4 times larger since 5 cows from the same farm gives less information than 5 cows from random farms.

Note. Cluster sampling is not as efficient as SRS. But, it's more convenient and cheaper in practice.

In general, say n is the total sample size. Then,

(1) SRS of n from N has variance $\left(1-\dfrac{n}{N}\right)\dfrac{\sigma^2}{n}$.

(2) SRS of $a = \frac{n}{B}$ clusters and obscure all B units in each sampled cluster has variance $\left(1-\dfrac{a}{A}\right)\dfrac{\sigma_A^2}{a}$.

Since n is the same,

$$\frac{n}{N} = \frac{aB}{AB} = \frac{a}{A}.$$

Then,

$$\frac{\mathrm{Var}\left(\tilde{\mu}_{\text{clu}}\right)}{\mathrm{Var}\left(\tilde{\mu}\right)} = \frac{\left(1-\frac{a}{A}\right)\frac{\sigma_A^2}{a}}{\left(1-\frac{n}{N}\right)\frac{\sigma^2}{n}} = \frac{B\sigma_A^2}{\sigma^2}.$$

In practice, units in the same cluster are similar, so σ_A^2 tends to be not much smaller than σ^2. So usually, cluster sampling has larger variance for the same n.

plot???

Lecture 11, 2025/10/08

Stratified sampling vs. cluster sampling:

- Stratified sampling works well if units within a stratum are similar among themselves. Samples from every stratum.

- Cluster sampling works poorly (i.e. high estimation variance) if units within a cluster are similar among themselves. Only get samples from some clusters.

Example. Suppose we have the following data.

Cluster	Unit Measurements	Cluster Mean
1	$y_{11}, y_{12}, \cdots, y_{1B}$	μ_1
2	$y_{21}, y_{22}, \cdots, y_{2B}$	μ_2
3	$y_{31}, y_{32}, \cdots, y_{3B}$	μ_3
$\vdots$	$\vdots$	$\vdots$
A	$y_{A1}, y_{A2}, \cdots, y_{AB}$	μ_A

Recall that the population variance of cluster means is

$$\sigma_A^2 = \frac{1}{A-1} \sum_{a=1}^{A} (\mu_a - \mu)^2,$$

which is different from the population variance of all units

$$\sigma^2 = \frac{1}{AB-1} \sum_{a=1}^{A} \sum_{b=1}^{B} (y_{ab} - \mu)^2.$$

Claim. If units within clusters are similar, then σ_A^2 will not be much smaller than σ^2.

Proof. Define

$$\sigma_B^2 = \underbrace{\frac{1}{A} \sum_{a=1}^{A} \underbrace{\left[\frac{1}{B-1} \sum_{b=1}^{B} (y_{ab} - \mu_a)^2 \right]}_{\text{pop. variance of cluster } a}}_{\text{average within-cluster variance}}.$$

Then, we can decompose σ^2 as follows:

$$\begin{aligned}
\sigma^2 &= \frac{1}{AB-1}\sum_{a=1}^{A}\sum_{b=1}^{B}(y_{ab}-\mu_a+\mu_a-\mu)^2 \\
&= \frac{1}{AB-1}\sum_{a=1}^{A}\sum_{b=1}^{B}\left[(y_{ab}-\mu_a)^2+(\mu_a-\mu)^2+2(y_{ab}-\mu_a)(\mu_a-\mu)\right] \\
&= \frac{1}{AB-1}\left[\sum_{a=1}^{A}\sum_{b=1}^{B}(y_{ab}-\mu_a)^2+B\sum_{a=1}^{A}(\mu_a-\mu)^2+2\sum_{a=1}^{A}(\mu_a-\mu)\underbrace{\sum_{b=1}^{B}(y_{ab}-\mu_a)}_{=B\mu_a-B\mu_a=0}\right]
\end{aligned}$$

Note that $B\mu_a$ is the cluster total of cluster a.

$$= \frac{1}{AB-1}\left[\sigma_B^2\cdot A(B-1)+\sigma_A^2\cdot B(A-1)\right].$$

So if A and B are large, then

$$\sigma^2 \approx \sigma_A^2+\sigma_B^2$$

i.e. population variance $\approx$ variance of cluster means + average within-cluster variance. In practice, σ_B^2 tends to be small, hence σ_A^2 will not be that small relative to σ^2. $\square$

Case 1: SRS of a out of A clusters measure all B units in each sampled cluster.

$$\hat{\mu}_{\text{clu}} = \frac{1}{a}\sum_{i\in S}\hat{\mu}_i, \quad \text{Var}\left(\tilde{\mu}_{\text{clu}}\right) = \left(1-\frac{a}{A}\right)\frac{\sigma_A^2}{a}.$$

Recall that

$$\frac{\text{Var}\left(\tilde{\mu}_{\text{clu}}\right)}{\text{Var}\left(\tilde{\mu}\right)} = \frac{B\sigma_A^2}{\sigma^2} \quad \text{often } > 1.$$

Case 2: Take SRS of a clusters out of A clusters, then take SRS of size b units out of B units in each sampled cluster. Overall sample size is $n = ab$. Now, instead of observing μ_i for sampled clusters i, we only get $\hat{\mu}_i$ from the sample mean of the b units out of B units in cluster i.

what's below?

$$\hat{\mu}_{\text{clu},2} = \frac{1}{a}\sum_{i\in S}\hat{\mu}_i,$$

$$\text{Var}\left(\tilde{\mu}_{\text{clu},2}\right) = \left(1-\frac{a}{A}\right)\frac{\hat{\sigma}_A^2}{a} + \left(1-\frac{b}{B}\right)\frac{\hat{\sigma}_B^2}{ab}.$$

The second term $\left(1-\frac{b}{B}\right)\frac{\hat{\sigma}_B^2}{ab}$ is the extra variance from having to estimate μ_i.

- If $b = B$, reduces to Case 1.
- In real situations, cluster sizes will not necessarily be equal (R survey package has formulas for that).

Analyzing a Cluster Sample in R

The survey package contains a California school performance dataset that we'll use as our example. The data is built-in so we can load it like this:

```
library(survey)
data(api)
```

There are two hypothetical cluster samples included:

- apiclus1: Take SRS of school districts and measure performance at all schools in sampled districts (case 1)
- apiclus2: Take SRS of school districts, then take SRS of schools within the sampled districts (case 2)

There are many variables in this dataset, let's focus on "college graduation percentage" (col.grad).

```
## dnum = district number (cluster)
## snum = school number (unit within cluster)
## Case 1
## tapply computes a function on a vector, by subgroup,
## e.g. cluster means:
tapply(apiclus1$col.grad,apiclus1$dnum,mean)
## fpc = number of districts
clus1 <- svydesign(id=~dnum, data=apiclus1, fpc=~fpc)
svymean(~col.grad, clus1)
```

```
# Case 2
## fpc1 = number of districts
## fpc2 = number of schools in that district
clus2 <- svydesign(id=~dnum+snum, data=apiclus2, fpc=~fpc1+fpc2)
svymean(~col.grad, clus2)
```